INCH-POUND

MIL-K-71129 (AR)
26 May 1992

MILITARY SPECIFICATION

KIT, REFURBISHMENT, FOR CARTRIDGE, 81MM, PRACTICE, SHORT RANGE, M880 LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for the refurbishment kit for Cartridge, 81MM Practice, Short Range, M880.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12944145 - Packing for Box, Fiberboard, Inner for Refurbishment Kit, M880
12944148 - Packing and Marking for Box, Fiberboard, Outer for Refurbishment Kit, M880
12944155 - Packing and Marking for Box, Wood, Wirebound for Refurbishment Kit, M880

(Copies of drawings and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications. The fuzes (M775), obturators, and ignition cartridges (M985) shall be from lots that meet their respective specification requirements and have been accepted by the Government.

3.2 Obturator. The obturator shall seat flush or below flush when properly assembled to a serviceable carrier round.

3.3 Breech plug firing pin position. The firing pin shall be flush to below flush relative to the rear face of the breech plug after assembly against the propelling charge.

3.4 Fuze striker arming indicator marking. The red arming indicator on the forward section of the fuze striker shall not be visible when the striker is in the safe position and shall be readily visible when the striker is in the armed position.

3.5 Increment plug. The increment plug shall be able to be easily inserted and be firmly seated in the increment plug hole of the projectile body, and be removable with an outward pull of 14 ± 8 pounds on the increment plug pull tab when pulled perpendicular to the projectile body.

3.6 Proving ground test. The refurbished cartridges shall comply with the following requirements when tested in accordance with 4.4.3.1 and 4.5.1.

3.6.1 Misfires. The refurbished cartridges shall not fail to fire from the weapon.

3.6.2 Short rounds. No refurbished cartridges shall have a horizontal range less than 80 percent of the average horizontal range (calculated without the short rounds).

3.6.3 Metal parts separation. There shall be no metal parts separation in the gun bore or in flight.

3.6.4 Range and deflection precision. The refurbished cartridge shall have a horizontal range probable error (PEr) less than 10 meters for all zones. The probable error deflection (PEd) shall be less than 10 meters for all zones.

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3.6.5 Duds. The fuze shall function upon ground impact of the refurbished cartridge.

3.6.6 Premature burst. There shall be no burst in the gun bore or in the first 15 meters of flight.

3.6.7 Weapon. The refurbished cartridge shall not be the cause of weapon damage.

3.6.8 Fuze signature. The fuze, when ballistically fired, shall provide both a visible and audible signature at a minimum range of 1500 meters from the observer. Only a visible or only an audible signature upon impact is not sufficient to meet this requirement. Visual and audio enhancement equipment shall not be used for this assessment. Visible is defined as flash or smoke or both flash and smoke.

3.7 First article . This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor are specified in the contract and by 4.3 herein.(See 6.2)

3.8 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks blisters, excess flash, corrosion products and any other foreign matter. Any cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The requirements of MIL-A-48078 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item.

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the components in the quantities indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

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4.3.3 Rejection. See MIL-A-48078 and the first article ballistic test shall be rejected if:

a. Two or more misfires occur.

b. Three or more duds occur without any misfires or if two or more duds occur with one misfire.

c. One or more short rounds occur.

d. One or more premature bursts occur.

e. One or more metal parts separation occur.

f. Precision requirements are not met. (70 degrees F conditioned samples only).

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-K-71129 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Packing for Box, Fiberboard, Inner for Refurbishment Kit, M880		See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Packing for Box, Fiberboard, Inner for Refurbishment Kit, M880 (dwg. 12944145)			
	Examination for defects	10	3.1	4.4.2.6
	Proving ground test	72	3.6	Test/4.5.1
	Plug, Increment (dwg. 12944200)			
	Examination for defects	10	3.1	4.4.2.5
	Increment plug pull test	10	3.5	Test/4.5.2
	Pin, Firing (dwg. 12944190)			
	Examination for defects	10	3.1	4.4.2.1

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Packing for Box, Fiberboard, Inner for Refurbishment Kit, M880	SHEET 2 OF 2	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Plug, Breech (dwg. 12944191) Examination for defects	10	3.1	4.4.2.2
	Breech Plug and Firing Pin Assembly (dwg. 12944192) Examination for defects	10	3.1	4.4.2.3
NOTES:				

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of kits shall contain:

a. Propelling charges made from not more than one lot of propellant, from one manufacturer.
b. Obturating rings of one lot interfix number, from one manufacturer.
c. Fuzes of one lot interfix, from one manufacturer.
d. Breech plug and firing pin assemblies from one lot interfix number, from one manufacturer.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the Classification of Characteristics paragraphs which follow. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes Sampling Inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs:

TABLE II. SAMPLING INSPECTION

LOT SIZE	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	13	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

(Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.)

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b. Alternative inspection provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.2). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the spoecified quality conformance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12944190
4.4.2.1	Pin, Firing		NEXT HIGHER ASSEMBLY 12944192

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total length	III	3.1	Gage
102	Large outside diameter	III	3.1	Gage
103	Small outside diameter	III	3.1	Gage
104	True position of small outside diameter to large outside diameter	III	3.1	Gage
105	Length of large outside diameter section	III	3.1	Gage
106	Length of firing pin from large outside diameter face	III	3.1	Gage
107	True position of firing point to large outside diameter	III	3.1	Gage
108	Firing point radius	III III	3.1 3.1	Gage Gage
Minor				
201	Improper surface finish	V	3.1	Visual
202	Protective finish missing or incomplete	V	3.1	Visual
203	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12944191
4.4.2.2	Plug, Breech		NEXT HIGHER ASSEMBLY 12944192

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Large inside diameter	III	3.1	Gage
102	Small inside diameter	III	3.1	Gage
103	Depth of large inside diameter	III	3.1	Gage
104	True position of large inside diameter to small inside diameter	III	3.1	Gage
105	True position of small inside diameter to outside thread	III	3.1	Gage
106	Outside thread major diameter, min.	III	3.1	Gage
107	Total length of plug	III	3.1	Gage
108	Pitch diameter of thread	III	3.1	Gage
Minor				
201	Outside diameter of plug	V	3.1	Gage
202	Length of outside diameter from front face to thread	V	3.1	Gage
203	Diameter of spanner wrench holes	V	3.1	Gage
204	Depth of spanner wrench holes	V	3.1	Gage
205	Location of spanner wrench holes	V	3.1	Gage
206	Protective finish missing or incomplete	V	3.1	Visual
207	Improper surface finish	V	3.1	Visual
208	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12944192
4.4.2.3	Breech Plug and Firing Pin Assembly		NEXT HIGHER ASSEMBLY 12944202

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Firing pin not free to move	100%	3.1	Manual
102	Stake protrusion inadequate	III	3.1	Gage
103	Firing pin does not exceed min. projection dimensions	III	3.1	Gage
Minor				
201	Poor workmanship	V	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Plug, Increment	1 OF 1	12944200
			NEXT HIGHER ASSEMBLY 12944202

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of plug insert	(a)	3.1	Gage
102	Length of plug insert	(a)	3.1	Gage
103	Thickness of plug tab	(a)	3.1	Gage
104	Width of pull tab	(a)	3.1	Gage
105	Thickness of pull tab to insert connector	(a)	3.1	Gage
106	Width of pull tab to insert connector	(a)	3.1	Gage
107	Increment plug pull test	V	3.3.5	Test/4.5.2
Minor				
201	Outside diameter of insert dome	(a)	3.1	Gage
202	Length of pull tab	(a)	3.1	Gage
203	Poor workmanship	V	3.8	Visual

NOTES: (a) 20 samples from each mold cavity shall be examined for this feature or dimension. The sample of 20 shall include the first two parts from the beginning of each production run and the last two parts from the end of the run. The remaining sample of (at least 16) shall be randomly selected during the production run. Any defects found in any of the 20 samples from each die cavity shall be cause for rejection of all increment plugs produced from that particular die cavity.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Packing and Marking for Box, Fiberboard, Inner for Refurbishment, Kit M880 (while packing)	1 OF 1	12944145
			NEXT HIGHER ASSEMBLY 12944148

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special A	Red marking visible on fuze striker	100%	3.1	Visual
Major 101	Kit components improperly packed	III	3.1	Visual
102	Kit components missing	III	3.1	Visual
Minor 201	Poor workmanship	V	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Packing and Marking for Box, Fiberboard, Inner for Refurbishment, Kit M880 (after packing)	1 OF 1	12944145
			NEXT HIGHER ASSEMBLY 12944148

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged	III	3.1	Visual
102	Contents loose	III	3.1	Visual/Manual/ Tactile
103	Sealing tape improperly applied or badly wrinkled	III	3.1	Visual
Minor				
201	Marking missing, misleading or unidentifiable	V	3.1	Visual
202	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Packing and Marking for Box, Fiberboard, Outer for Refurbishment, Kit M880 (while packing)	1 OF 1	12944148
			NEXT HIGHER ASSEMBLY 12944155

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Wrong number of kits in box	III	3.1	Visual
102	Tape improperly applied to box	III	3.1	Visual
103	Barrier bag in properly placed or sealed	III	3.1	Visual
Minor				
201	Marking missing, misleading or unidentifiable	V	3.1	Visual
202	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Packing and Marking for Box, Wood, Wirebound for Refurbishment, Kit M880 (while packing)	1 OF 1	12944155
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Damaged barrier bags	III	3.1	Visual
Minor 201	Spacers or fillers inadequate to prohibit movement of contents	V	3.1	Manual/Tactile
202	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Packing for Box, Wood, Wirebound for Refurbishment, Kit M880 (after packing)		12944155
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Boards split or loose, exposing box's contents	V	3.1	Visual
202	Box's contents loose	V	3.1	Visual/Tactile/Manual
203	Marking missing, misleading or unidentifiable	V	3.1	Visual
204	Box wire locking tabs insecure	V	3.1	Visual/Manual
205	Seal missing, unsealed or improperly positioned	V	3.1	Visual/Manual
206	Poor workmanship	V	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Packing and Marking for Box, Fiberboard, Inner for Refurbishment, Kit M880 (after refurbishing a spent cartridge)	1 OF 1	12944145
			NEXT HIGHER ASSEMBLY 12944148

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Red marking visible on fuze striker	(a)	3.1	Visual
B	Fuze striker in armed position	(a)	3.1	Visual
C	Fuze packing clip missing or improperly assembled	(a)	3.1	Visual
D	Obturating ring missing	(a)	3.1	Visual
E	Breech plug assembly firing pin not below flush after assembly to spent cartridge	(a)	3.1	Visual
F	Powder from energetic cartridges found on outside of round due to poor seals on shotgun shells	(a)	3.1	Visual
Major				
101	Obturating ring not flush or below bourrelet diameter	(a)	3.2	Gage
102	Firing pin does not move freely within breech plug	(a)	3.1	Manual/Tactile
103	Increment plug pull test	III	3.5	Test/4.5.2
104	Proving ground test	See4.4.3.1 (b)	3.6	Test/4.5.1
Minor				
201	Poor workmanship	V	3.8	Visual

NOTES: (a) All test quantities.
(b) The proving ground test shall not be performed prior to the successful completion of all examinations for defects and testing.

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4.4.3 Testing.

4.4.3.1 Proving ground test. The cartridges shall be observed for the following characteristics:

a. Misfires, major defect (see 3.6.1)
b. Duds, major defect (see 3.6.5)
c. Short rounds, special defect (see 3.6.2)
d. Premature burst, critical defect (see 3.6.6)
e. Metal parts separation, special defect (see 3.6.3)
f. Precision, major defect (see 3.6.4)
g. Weapon damage, major defect (see 3.6.7)
h. Proper impact signature, major defect (see 3.6.8)

4.4.3.1.1 Observation criteria.

4.4.3.1.2 Sampling and acceptance criteria. Testing shall be as specified in 4.5.1. The lot shall be rejected if in sections a, b and c of 4.5.1.1 combined:

a. Two or more misfires occur.
b. Three or more duds occur without any misfires or if two or more duds occur with one misfire.
c. One or more short rounds occur.
d. One or more premature bursts occur.
e. One or more metal parts separation occur.
f. Precision requirements are not met, at ambient temperature (70 degrees F).
g. If the weapon is damaged by abnormal wear of firing pin, or caused by excessive pressure.
h. Two or more improper impact signatures.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See section 6 of MIL-A-48078 and 6.2 herein.

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4.5 Methods of inspection.

4.5.1 Proving ground ballistic test.

4.5.1.1 Test group and firing criteria. The test shall be performed at a Government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The refurbished cartridges shall be fired from a standard 81MM M252 mortar, with M3 series baseplate, at an elevation of forty-five (45) degrees of 800 mils. The refurbished cartridges shall be grouped as follows:

a. One group of 20 each conditioned for at least 24 hours at +110 degrees F ± 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 3.

b. Two groups of 16 each conditioned for at least 24 hours at 70 degrees F ± 2.5 degrees F (see 4.4.3.1.2), one group shall be fired at charge 0 and the other at charge 3.

c. One group of 20 each conditioned for at least 24 hours at 0 degrees F ± 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 0.

All of the above refurbished cartridges shall be fired within two minutes after removal from the conditioning chamber. A smear camera shall be used to photograph all samples. The camera shall be positioned to pick up each round just out of the gun muzzle. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all other rounds in that particular charge zone, in that lot, shall be attached to the firing record. Film from acceptable lots may be discarded. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. Film from acceptable lots may be discarded. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record.

The individual muzzle velocities, deflections and ranges shall be recorded on the firing record. Range probable error shall be calculated and recorded on the firing record for groups a, b and c above.

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4.5.1.2 Determination of safe fuze condition.

4.5.1.2.1 External fuze examination. Before removal of the ogive, the red band on the striker which indicates an armed fuze must not be visible. Similarly, loose powder from the spotting cartridge must not be visible. If either or both of these conditions are found, the fuze shall be considered unsafe to handle.

4.5.1.2.2 Damaged fuze. A broken fuze shall be considered safe to dispose of providing that the spotting cartridge is still intact and the striker half of the fuze can no longer come in contact with the spotting cartridge.

4.5.2 Increment plug, pull test. The increment plugs shall be removed from the projectile body by a direct pull normal to the projectile surface and gaged by pull measuring devices simultaneously. Observation shall be made for the force required to remove the plug. If any of the test sample units fail to meet the pull requirement limits, that sample unit shall be classified defective. As an alternative method, an approved gaged can be substituted for the projectile body.

4.5.3 Test validity. If for any reason the ARDEC considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

5. PACKAGING

5.1 Unit packing.

5.1.1 Level A. Unit packing for refurbishment kit shall be in accordance with requirements of drawing 12944145.

5.1.2 Level B. Same as Level A.

5.1.3 Level C. Same as Level A.

5.2 Intermediate packing.

5.2.1 Level A. Intermediate packing for refurbishment kits shall be in accordance with requirements of drawing 12944148.

5.2.2 Level B. Same as Level A.

5.2.3 Level C. Same as Level A.

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5.3 Packing.

5.3.1 Level A. Packing for refurbishment kits shall be in accordance with requirements of drawing 12944155.

5.3.2 Level B. Same as Level A.

5.3.3 Level C. Same as Level A.

5.4 Marking.

5.4.1 Level A. Unit packing for refurbishment kits shall be marked in accordance with the requirements of drawing 12944145. Intermediate packing for refurbishment kits shall be marked in accordance with the requirements of drawing 12944148. Packing for refurbishment kits shall be marked in accordance with the requirements of drawing 12944155.

5.4.2 Level B. Same as Level A.

5.4.3 Level C. Same as Level A.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The item covered by this specification is intended for use in Cartridge, 81mm, Practice Short Range, M880.

6.2 Acquisition requirements.

a. See MIL-A-48078.

b. Issue of DODSS to be cited in the solicitation and if required, the specific issue of individual documents referenced (See 2.1).

c. Provisions for submission of first article sample.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied to the contract.

Reference Paragraph	DID Number	DID Title
4.5.3.3.3	DI-S-3628	Nondestructive Inspection Plan

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6.4 Submission of inspection equipment designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, AMCCOM, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to the following: Commander, ARDEC, ATTN: SMCAR-QAT-B, SMCAR-FSS-DM, and SMCAR-FSA-MM, Picatinny Arsenal, NJ 07806-5000.

6.7 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.8. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.9 Subject term (key word) listing.

Ballistic tests
Short range
Deflection precision
Pull test
Workmanship criteria

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AD89)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisio letter should be given.
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1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
MIL-K-71129(AR)	26 MAY 1992

3. DOCUMENT TITLE KIT, REFURBISHMENT, FOR CARTRIDGE, 81MM, PRACTICE, SHORT RANGE, M880 LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Paul Tremblay	(1) Commercial 201-724-6671 (2) AUTOVON DSN-880-6671
c. ADDRESS (Include Zip Code) Commander: U.S. Army ARDEC ATTN: SMCAR-BAC-S Picatinny Arsenal, New Jersey 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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INCH-POUND

MIL-K-71129 (AR)
AMENDMENT 3
23 April 1996
SUPERSEDING
AMENDMENT 2
05 December 1995

MILITARY SPECIFICATION

KIT, REFURBISHMENT, FOR CARTRIDGE, 81MM
PRACTICE, SHORT RANGE, M880
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-K-71129 (AR), dated 26 May 1992, and is approved for use by the U.S. Army Armaments Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3: Delete: "propelling charge", and substitute: "ignition cartridge".

Delete 3.6.1 in its entirety and substitute:

"3.6.1 Misfires. The cartridges shall not fail to fire from the weapon when they are initially dropped into it."

Delete 3.6.4 in its entirety and substitute:

"3.6.4 Range and deflection precision. The cartridge shall have a horizontal range probable error (PEr) less than 15 meters, for all zones when calculated to the nearest one tenth (0.1) meter. The deflection probable error (PEd) shall be less than 15 meters for all zones when calculated to the nearest one tenth (0.1) meter."

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3.6.6: Add: "unobstructed", before the word "flight".

3.6.9 Add the following:

3.6.9 Dud plug function.

"3.6.9.1 Functioned round. All dud plug heads (regardless of stems) shall blow out of the projectile body upon fuze function."

"3.6.9.2 Inert round. No dud plug head (regardless of stems) shall exit the projectile body during ballistic flight or upon impact."

3.9 Add the following:

"3.9 Dud plugs."

"3.9.1 Configuration. Three dud plugs shall be present and properly inserted into the projectile body for the proving ground test."

"3.9.2 Dud plug push in - push out force. No dud plug shall push into the required test plate using a force greater than 40.0 pounds and shall not be capable of being pushed out of the required test plate using a force of less than 25.0 pounds at a temperature of 70 degrees F ± -3 degrees F when tested as specified in 4.5.4."

4.3.3: Delete "g" and "h" in their entirety and substitute:

"g. If the weapon is damaged by abnormal wear of the interior or of the firing pin, or caused by excessive pressure."

"h. Two or more improper impact signatures occur."

Add the following:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test."

"j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

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PAGE 7

Table I: Proving ground test line, under Conformance Criteria:

Delete: "72", and substitute: "96".

Add the following:

"Dud Plug ...(dwg. 12961001)

Examination for defects ...30 ... 3.1 ... 4.4.2.11

Push in - push out test ... 30 ... 3.9.2 ... 4.5.4".

PAGE 9

4.4.1: Delete the word "interfix" from the text from lines a., b., c., and d.

Add the following:

"e. Dud plugs from one inspection lot that comply with 3.9.2."

PAGE 11

4.4.2.1, Major 105: Change call out:

from: "Length of large outside diameter section",

to: "Length between large outside diameter aft face to firing pin point".

PAGE 15

4.4.2.5: Delete Special A in its entirety.

Critical: Add the following:

"1 Fuze striker in armed position ... 100% ... 3.1 ... Visual".

PAGE 16

4.4.2.6: Delete Major 103 in its entirety.

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* 4.4.2.10: Delete Specials A, B, C, D, E, F and G in their entirety and add the following:

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
"1	Fuze striker in armed position	(a)	3.1	Gage
2	Breech plug assembly firing pin not below flush after assembly to round	(a)	3.3	Gage
3	Powder from energetic cartridges found on outside of round	(a)	3.1	Visual
Major				
105	Red marking visible on fuze striker	(a)	3.4	Visual
106	Fuze packing clip missing or improperly assembled	(a)	3.1	Visual
107	Obturator ring missing	(a)	3.1	Visual
108	Any dud plugs missing	(a)	3.9.1	Visual"

PAGE 21

4.4.2.11: Add the following:

PARAGRAPH	TITLE			DRAWING NUMBER
"4.4.2.11	Dud Plug	SHEET 1	OF 1	12961001
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Push in - push out test	30	3.9.2	Test/4.5.4
Minor				
201	Poor workmanship	30	3.6	Visual"

4.4.3.1: Delete in its entirety and substitute:

"4.4.3.1 Proving ground test."

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4.4.3.1.1, Add:

"The refurbished cartridges shall be observed for the following characteristics:

a. Misfires, major defect (see 3.6.1)
b. Duds, major defect (see 3.6.5)
* c. Short rounds, major defect (see 3.6.2)
d. Premature burst, critical defect (see 3.6.6)
* e. Metal parts separation, major defect (see 3.6.3)
f. Precision, major defect (see 3.6.4)
g. Weapon damage, major defect (see 3.6.7)
h. Improper impact signature, major defect (see 3.6.8)
i. All dud plug heads out of any one projectile in the inert fuze ballistic test, Critical defect (see 3.6.9.2)
j. Two or more dud plug heads remain in the live fuze ballistic test, Major defect (see 3.6.9.1)".

4.4.3.1.2, Add:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

Add the following:

"d. One group of 24 each with inert fuzes conditioned for at least 24 hours at 0 degrees F +/- 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 3."

4.5.1.1, last paragraph: Delete and substitute:

"The individual deflections and ranges (accuracy of both to 0.01 meter) and muzzle velocities shall be calculated to the nearest 0.1 meter and recorded on the firing record for groups a, b, and c above. All rounds shall be fired onto a flat, nongrassy earthen surface that is not marshy or frozen."

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Add the following:

"4.5.4: Push in - push out test. Major defect. A hardened steel test plate of 1 x 2 x 1/4 inch is required for this test. There is to be a 0.250 ± .001 diameter thru hole (having an internal surface finish of 32 maximum) in the steel plate. The sample of dud plugs to be tested shall be preconditioned at a temperature of 100 degrees F ± 5 degrees F for a period of not less than 24 hours nor more than 48 hours prior to being subjected to this test. The preconditioned dud plugs shall be tested not earlier than 15 minutes after removal from the conditioning box nor later than one (1) hour after removal from the conditioning box. Press a sample dud plug into the hole using an arbor press until the head is 0.0625 ± 0.02 inch above flush with the plate surface. At this point, the push in test is completed. Record the maximum force. Turn the steel plate over and push the dud plug out of the steel plate to 0.0625 ± 0.02 inch above flush using the same arbor press. Record the maximum force required to push out the dud plug. At this point, the push out test is completed. Compute sample mean (xbar) and standard deviation (sd). Insert these into the following formulas:

Accept if:

a. xbar(In) = 37.0 pounds max., xbar(In) + 3sd(In) = 40.0 pounds max.

and

b. xbar(Out) = 30.0 pounds min., xbar(Out) - 3sd(Out) = 25.0 pounds min. and no individual x(Out) less than 25.0 pounds.

Any dud plug sample that does not meet the above requirements shall be cause for lot rejection.

Each test sample quantity may be examined for outliers (high side only) using the Grubbs T statistical test (see 6.12). A maximum of two outliers may be removed from the sample quantity."

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6.10 Delete in its entirety.

6.11 Add the following:

"6.11 Inspection lot definition. A homogeneous collection of units of product or one shipment of component parts or assemblies from a subcontractor, bearing identification and treated as a unique entity from which a sample is to be drawn and inspected (and tested) to determine conformance with the acceptability criteria."

6.12: Add the following:

"6.12 Outlier test. When the accept criteria is not met and if it is suspected that a statistical outlier may be present in the sample of 30 observations. Arrange the observed values in order of increasing magnitude (X1, X2 .. Xn-1, Xn. Using the statistics calculated for the sample, calculate Tn where:

If X1, the smallest value, rather than Xn, is the doubtful value the test criterion is:

$$Tn = \frac{Xn - \bar{X}}{s}$$

If X1, the smallest value, rather than Xn, is the doubtful value the test criterion is:

$$T1 = \frac{\bar{X} - X1}{s}$$

Where:

$\bar{X}$ = arithmetic average of all n values
s = estimate of the population standard deviation based on the sample data calculated as follows:

$$s = \left[\frac{\sum (x_i - \bar{x})^{1/2}}{n-1}\right] = \left[\frac{n\sum x_i^2 - \left(\sum x_i\right)^2}{n(n-1)}\right]^{1/2}$$

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If Tn is less than or equal to 2.745, the highest value may not be eliminated. The lot is rejected. If Tn is greater than or equal to 2.745, the largest value obtained in the sample shall be eliminated and the sample statistics X and s shall be recalculated using the twenty-nine remaining values. This shall be compared to the accept criteria. If the criteria is met, accept the lot. If not, the same procedure shall be followed for the Xn-1 observation. Calculate Tn-1 based on the new statistics of the remaining sample of 29. If Tn-1 is less than or equal to 2.730, the lot is rejected. If Tn-1 is greater than or equal to 2.730, then this observation shall be eliminated and the sample statistics calculated based on the remaining 28 observations. If these values meet the accept criteria, accept the lot. If the accept criteria is still not met, reject the lot."

6.13: Add the following:

"6.13 Marshy ground. Marshy ground is defined as an earthen surface that contains standing water or free flowing (fluid) mud or a combination of both.

6.14 Add the following:

"6.14 Frozen ground. Frozen ground is defined as an earthen ground within which water has frozen creating a hard surface."

The margins of this amendment are marked with an asterisk, or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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